BORNEOHACKWND 2026

PROJECT REPORT
THEME : TECH FOR IMPACT
PROJECT TITLE : AI-POWERED FOOD REDISTRIBUTION PLATFORM

Sustainable Development Goals :

SDG 2 : Zero Hunger

SDG 12: Responsible Consumption And Production

SDG 13: Climate Action

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1.0 Introduction

Background of Case Study

Food security in ASEAN is challenged by climate change, supply chain inefficiencies, and rapid urbanization. While millions of people face food shortages, large amounts of edible food are wasted daily. This imbalance highlights the need for technology-driven solutions to improve food distribution and reduce waste.

Chosen SDG

- SDG 2 – Zero Hunger

Improve food accessibility for vulnerable communities.

- SDG 12 – Responsible Consumption and Production

Reduce food waste through efficient redistribution.

- SDG 13 – Climate Action

Reduce greenhouse gas emissions by minimizing food waste and promoting sustainable food distribution

Problem Statement:

Despite the availability of surplus food from businesses and restaurants, many NGOs and communities still struggle to access sufficient food due to inefficient coordination and lack of real-time distribution systems.

Objectives:

- Reduce food waste through surplus food redistribution
- Connect donors with NGOs using a digital platform
- Improve efficiency of food distribution with AI matching
- Support sustainable food systems in urban communities

2.0 Review of Existing Solutions

2.1 Existing Solutions

- Food rescue platforms (e.g., food banks, donation apps)
- Manual NGO coordination for food distribution
- Community-based food sharing initiatives

2.2 Limitations

- Lack of real-time matching between donors and NGOs
- Inefficient coordination causing delays
- Limited use of AI or smart location-based distribution
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2.3 Sparebite Solution:

AI-powered platform that automatically connects surplus food donors with nearby NGOs.

3.0 Methodology

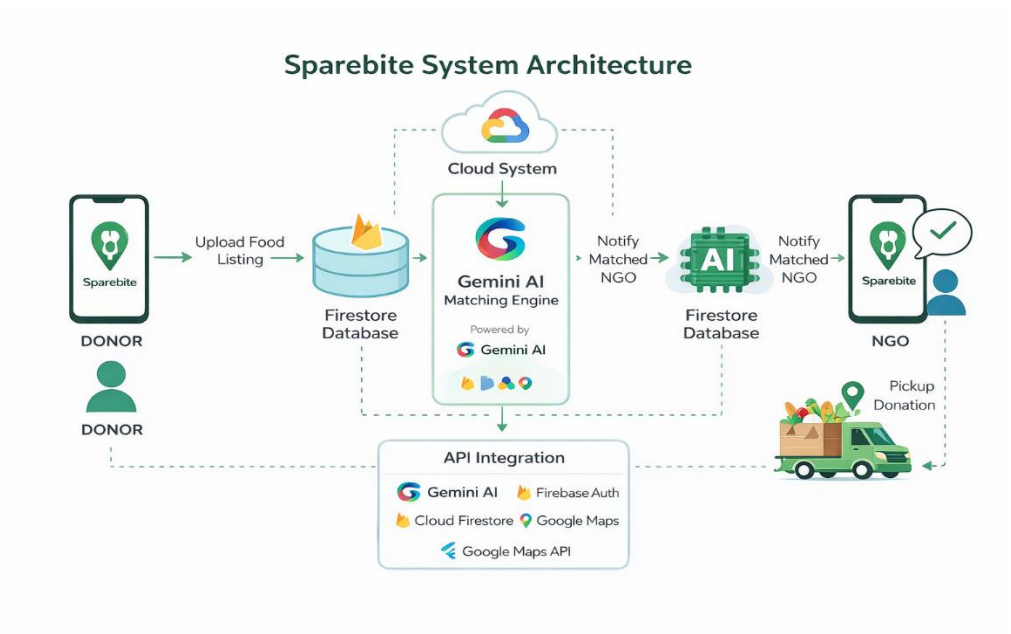
3.1 Technology Stack

- **Flutter** – Cross-platform mobile application development
- **Firebase Authentication** – Secure user login and registration
- **Cloud Firestore** – Real-time database for storing food listings and user data
- **Google Maps API** – Location-based matching between donors and NGOs
- **Gemini AI** – Intelligent matching engine for food distribution

3.2 System Architecture

The Sparebite system connects food donors with NGOs through a cloud-based architecture.

- Donors upload surplus food listings through the mobile application.
- The data is stored in **Cloud Firestore**.
- **Gemini AI** analyzes factors such as location, availability, and NGO status.
- The system identifies the most suitable NGO.
- The NGO receives a notification and can accept or decline the donation.



3.3 Implementation Process

- Designed user interface flows for donors and NGOs
- Developed food listing and management features
- Integrated Firebase services for authentication and database
- Implemented Gemini AI matching logic
- Conducted user testing and feedback collection

4.0 Results

- Successfully developed a working prototype of Sparebite
- Achieved efficient food donation workflow
- Reduced coordination gap between donors and NGOs

4.2 User Feedback

- 15 beta testers participated
- 100% found the app easy to understand
- Positive feedback on AI matching concept

5.0 Prototype Showcase

Key Features

- Donor Dashboard
- Upload Food Listing
- My Listings Management
- NGO Dashboard & Notification System

6.0 AI Acknowledgement (Mandatory)

AI Tools Used

- Notebooklm - for slides design
- ChatGPT - assistance in documentation, idea refinement, and debugging guidance
- Cursor – IDE assistant that helps wih coding
- AI design tools – generation of presentation visuals and UI concepts

Usage
AI tools were used to support development, improve documentation clarity, and generate design ideas during the project.

7.0 Conclusion

Sparebite provides a technology-driven solution to reduce food waste and improve food accessibility. By connecting surplus food donors with NGOs through AI-powered matching, the platform supports sustainable food distribution and contributes to achieving SDG 2, SDG 12 and SDG 13.

Future Improvements

• Expand donation sources to include grocery stores, food factories, and farms.
• Integrate predictive AI to improve food–NGO matching accuracy.

- Add logistics and route optimization for efficient food pickup and delivery.
- Develop a volunteer network to support food distribution.
- Expand the platform to support more NGOs and cities.